Somnath Rakshit

somnath52@gmail.com | somnath.pages.dev | linkedin.com/in/somnathrakshit/ | github.com/somnathrakshit/

RELEVANT EXPERIENCE

Senior Data Scientist, Walmart Global Tech, Bentonville, AR — July 2021 – Ongoing

- Architected an autonomous **agentic AI** system to automate support manager workflows by processing **unstructured data** from associate emails.
- Integrated **data reasoning models** with a **RAG** framework, enabling the agent to perform multi-step actions and autonomously request user clarifications.
- Architected and deployed robust **ML infrastructure** using FastAPI, Docker, Kubernetes and CI/CD pipelines for model training and inference
- Implemented automated **data quality checks** and **validation pipelines**, and comprehensive **monitoring and observability** systems for ML pipeline performance, achieving 99.9% system availability
- Designed **auto-scaling** and **load balancing solutions** handling thousands of daily requests across distributed systems

Software Engineer – Machine Learning, Cyware Labs — July 2018 – Nov 2018

- Deployed ML models on AWS EC2 using cloud-native technologies including Celery and Gunicorn
- Developed and implemented CNN-based clustering system improving article selection efficiency by 100%
- Built scalable NLP pipelines for entity recognition and trend analysis

EDUCATION

Master of Science, Information Studies (Data Science Track) — May 2021
The University of Texas at Austin *(GPA: 3.85/4)*

Relevant Courses: Data Mining, Introduction to Machine Learning, Linear Models, Mathematical Statistics for Applications.

Bachelor of Technology, Computer Science and Engineering — May 2018
Jalpaiguri Government Engineering College, India *(GPA: 8.68/10)*

TECHNICAL SKILLS

Programming: Python, Java
ML & GenAI: PyTorch, Hugging Face Transformers, RAG, Finetuning, LangChain
MLOps & CI/CD: Docker, Kubernetes, Jenkins, FastAPI
Data & Cloud: Pandas, Numpy, SQL, Databricks, Milvus, GCP

RELEVANT PROJECTS

A GPU-accelerated MRI Sequence Simulation for Differentiable Optimization and Learning *Guide: Dr. Jon Tamir* — Mar-May 2020

- Wrote an open source GPU accelerated simulator in **PyTorch** for the Extended Phase Graph Algorithm for sets of sequences. After successfully parallelizing the simulator, we increased its speed by almost 100,000 times. This work was accepted as an abstract in ISMRM 2021 virtual conference.

geograpy3 — Sep 2018 (Ongoing)

- geograpy3 is a Python library that is used to extract place names from a URL or text, and add context to those names, e.g., distinguishing between a country, region or city. It can be installed from PyPi using pip and has over 245,000 downloads.